Shivam Chaturvedi

PhD Candidate

Indian Institute of Science, Bangalore

+91 9027667516 | shivamchatur@iisc.ac.in | imshivam24.github.io | shivam-che

Education

Indian Institute of Science *Bangalore*

PhD Chemical Engineering *2022 - present*

- Advisor: Prof. Ananth Govind Rajan

Malaviya National Institute of Technology *Jaipur*

B.Tech Chemical Engineering *2018 - 2022*

- Thesis research advisor: Prof. Madhu Agarwal

Publications

Published

Shivam Chaturvedi, Abhijit Gogoi, and Ananth Govind Rajan, *Hydrocarbon Selectivity in CO and CO_2 Reduction on Al–Cu Single-Atom Alloy Electrocatalysts: Effects of Potential, pH, and Facet*, **ChemSusChem**, 19, e202501817 (2026).

Shivam Chaturvedi, Amar Deep Pathak, Nishant Sinha, and Ananth Govind Rajan, *Transient Microkinetic Modeling of Electrochemical Reactions: Capturing Unsteady Dynamics of CO Reduction and Oxygen Evolution*, **Advanced Theory and Simulations**, 9, e00799 (2025).

Raghavendra Rajagopalan*, **Shivam Chaturvedi***, Neeru Chaudhary*, Abhijit Gogoi, Tej S Choksi, Ananth Govind Rajan, *Advances in CO_2 reduction on bulk and two-dimensional electrocatalysts: From first principles to experimental outcomes*, **Current Opinion in Electrochemistry**, 51, 101668 (2025).

Bivas Bhaumik, **Shivam Chaturvedi**, Satyasaran Changdar, Soumen De, *A unique physics-aided deep learning model for predicting viscosity of nanofluids*, **International Journal for Computational Methods in Engineering Science and Mechanics**, 24, 167–181 (2022).

In Press

Anand Mohan Verma*, **Shivam Chaturvedi***, Swastik Paul*, Ananth Govind Rajan and Others, *Data-driven massive reaction networks reveal new pathways underlying catalytic CO_2 hydrogenation*, **Nature Communications** (2026)

In Preparation

Shivam Chaturvedi, Nitish Govindarajan, Ananth Govind Rajan, *Copper's Missing Piece: A Machine-Learning and Graph-Theoretical Screening of Cu Single-Atom Alloys for CO_2 Reduction*

Anoop P. Pushkar*,**Shivam Chaturvedi***, Ananth Govind Rajan, *Hydronium and Water Proton-Transfer Pathways Together Govern pH-Dependent Selectivity in Electrochemical CO_2 Reductionn*

* These authors contributed equally

Presentations

Oral Talks

Mar 2026. *Machine learning-guided design of copper-based single-atom alloy catalysts for CO_2 valorization (Online)* at the ACS Spring'26.

Jan 2026. *Advancing CO_2 Conversion via Data-Driven Pathways* at the IISc Chemical Engineering Symposium (ICS-26).

Jan 2026. *Unraveling Hidden Reaction Pathways in CO_2 Hydrogenation Using Machine Learning (Online)* at IIPE, Visakhapatnam.

Jan 2026. *Advancing CO_2 Conversion via Data-Driven Pathways* at the Faculty Development Programme (FDP), IISc, Bangalore.

Nov 2025. *Transient Microkinetic Modeling of Electrochemical Mechanisms: Towards Understanding the Dynamics of CO Reduction and Oxygen Evolution* at the AIChE Annual Meeting, Boston, Massachusetts.

Nov 2025. *Data-Driven Massive Reaction Networks Reveal New Pathways Underlying Catalytic CO_2 Hydrogenation* at the AIChE Annual Meeting, Boston, Massachusetts.

Oct 2023. *Microkinetic Modelling for Electrochemical Systems* at the Shell.ai Scientific Conference, Bangalore.

July 2023. *AI/ML for 2D Materials* at Namma Psi-k, JNCASR, Bangalore.

Poster Presentations

Dec 2025. *Transient Microkinetic Modeling of Electrochemical Mechanisms: Towards Understanding the Dynamics of CO Reduction and Oxygen Evolution* at ICCCU25, JNCASR, Bangalore.

Sep 2024. *Copper's Missing Piece: Screening Copper-Based Single-Atom Alloys for Electrochemical CO_2 Reduction Using Machine Learning* at the CARE Symposium, IIT Madras.

May 2024. *Copper's Missing Piece: Screening Copper-Based Single-Atom Alloys for Electrochemical CO_2 Reduction to Propylene* at the 2nd Divisional Research Symposium, IISc, Bangalore.

Oct 2023. *Development of a Microkinetic Framework for Electrochemical CO_2 Reduction* at the Shell.ai Scientific Conference, Bangalore.

Teaching Experience

Aug 2026	**NPTEL: Machine Learning for Core Engineering Disciplines**, Teaching Assistant under Prof. Ananth Govind Rajan
Aug 2025	**NPTEL: Machine Learning for Core Engineering Disciplines**, Teaching Assistant under Prof. Ananth Govind Rajan
Jan 2025	**CH253: Quantum-Mechanical Modelling of Nanomaterials**, Teaching Assistant under Prof. Ananth Govind Rajan
Aug 2024	**CH202: Numerical Methods**, Teaching Assistant under Prof. Narendra Dixit

Research Experience

Shell Technology Centre (STC) *Bangalore, KA*

Co-Advisors: Dr. Amardeep Pathak, Dr. Nishant Sinha, and Prof. Ananth Govind Rajan *Apr. 2023 – Oct. 2024*

Unsteady-State Microkinetic Modeling of Electrochemical CO_2 Reduction

- Developed and tested a wrapper for MKMCXX, optimizing it for electrochemical CO_2 reduction simulations
- Built microkinetic models for reaction pathways and evaluated catalyst performance

Malaviya National Institute of Technology *Jaipur, RJ*

Advisor: Dr. Madhu Agarwal *Aug. 2021 – Mar. 2022*

Development of a Bio-Adsorbent for Arsenic Removal Using a Hydrothermal Process

- Developed orange peel–based bio-adsorbents for arsenic removal from potable water in rural India
- Conducted adsorption studies and characterized material performance

IASc–INSA–NASI Summer Research Fellowship Programme *Remote (Online)*

Advisor: Dr. Venkadachalam Ramesh (Central University of Tamil Nadu) *June 2020 – July 2020*

Analyzing COVID-19 Spread in Chest X-rays Using CNNs

- Designed a CNN-based X-ray analysis model to predict COVID-19 with 96.9% precision and 91.7% recall
- Utilized ResNet with transfer learning for rapid and cost-effective disease detection

Institute of Technology & Management, Salt Lake *Remote (Online)*

SUPERVISOR: DR. SATYASARAN CHANGDAR *Aug. 2020 – Feb. 2021*

Predicting the Viscosity of Various Nanofluids Using Artificial Neural Networks (ANNs)

- Explored a physics-guided neural network framework to develop a generalized model capable of predicting nanofluid viscosity with high accuracy

Outreach

LEADERSHIP & SERVICE

2024-2025	**Chemical Engineering Association**, Vice President	*IISc*
2023-2025	**Hockey Club**, Club Convenor	*IISc*
2024-2025	**Institute Innovation Council**, Innovation Coordinator	*IISc*
2020-2021	**NSS (National Service Scheme)**, Core Team Member	*MNIT*
2018-2019	**Energy Club**, Executive Member	*MNIT*